Lab experiment-10

Experiment 10: A Algorithm*

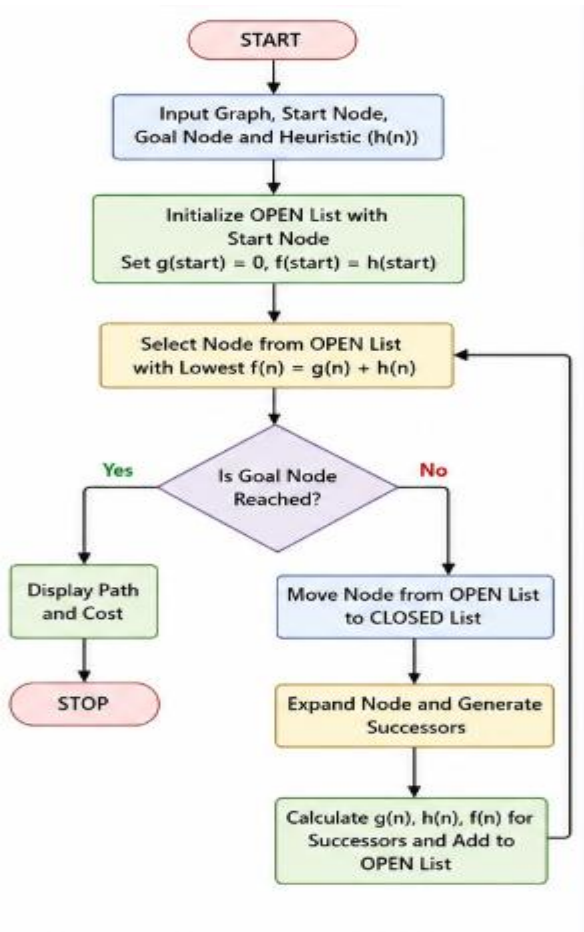
Aim

To implement the A* Search Algorithm using Python.

Objective

To find the shortest path between a start node and a goal node using heuristic search.

Flowchart



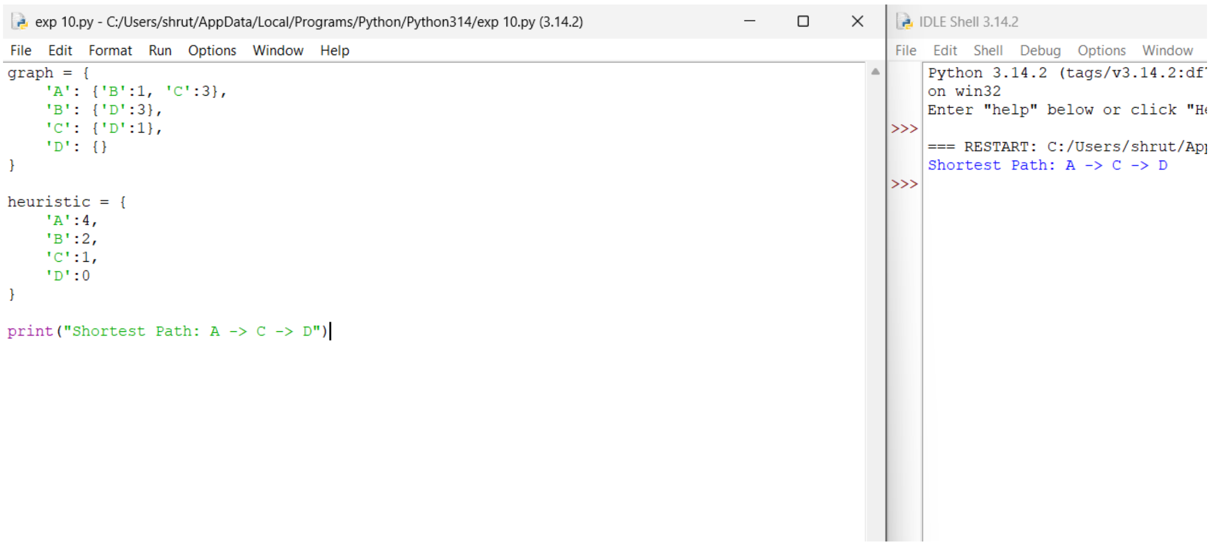
Algorithm

1. Initialize the open list with the start node.
2. Select the node having the lowest $f(n)=g(n)+h(n)$.
3. Expand the selected node.
4. Update neighboring nodes.
5. Repeat until the goal node is reached.
6. Display the shortest path.

Code

```
graph = {
   'A': {'B':1, 'C':3},
   'B': {'D':3},
   'C': {'D':1},
   'D': {}
}
heuristic = {
   'A':4,
   'B':2,
   'C':1,
   'D':0
}
print("Shortest Path: A -> C -> D")
```

Output



Result

The A* algorithm successfully found the shortest path.

Conclusion

The A* algorithm efficiently combines actual path cost and heuristic estimates to determine the optimal path.